Cloud-Based Behavioral Insider Threat Prediction Using Explainable AI (XAI)

Course: BITE412L - Cloud Computing - Project Phase-I Review Report

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1 Abstract

Insider threats are a serious security challenge because malicious or negligent users operate with legitimate credentials and can bypass traditional perimeter controls. Firewalls, antivirus tools and signature-based intrusion detection systems are not sufficient for identifying subtle activities such as **off-hour access, abnormal file copying, external email transfer, suspicious USB usage and long-term behavioral drift.** Recent research shows that **machine learning and deep learning models** can improve detection, but many approaches still face class imbalance, limited real-time deployment, weak integration of heterogeneous logs and poor explainability for Security Operations Center analysts.

This project proposes a cloud-based behavioral insider threat prediction framework using Explainable AI. The system uses the **CMU CERT Insider Threat Dataset r4.2** as the benchmark source and processes logon, file, device, email and web activity logs into user-session features. A hybrid prediction layer combines ensemble learning models such as **XGBoost** with temporal models such as **BiLSTM** to detect both isolated anomalies and sequential behavior changes. **SHAP-based explainability** is used to show why a user session was flagged, helping analysts understand the contribution of features such as after-hours login frequency, removable media use and sensitive file access.

The proposed solution is planned on **Microsoft Azure using Azure Blob Storage, Azure Data Factory, Azure Functions, Azure Machine Learning, Azure SQL Database or Cosmos DB, Microsoft Entra ID, Azure Monitor, Azure Notification Hubs and Power BI.** The framework aims to deliver scalable log processing, interpretable risk scoring, secure storage, automated alerts and dashboard-based monitoring for practical insider threat investigation.

2 Literature Survey and Individual Research Gap Analysis

The selected papers focus on insider threat detection, explainable AI, temporal behavior modeling, ensemble learning and cloud-ready security analytics. The literature survey below summarizes the core method, dataset, strengths, limitations and research gap for each assigned paper.

Paper	Method	Dataset	Advantages	Limitations	Research Gap
Beyond Firewall: Leveraging Machine Learning for Real-Time Insider Threats Identification and User Profiling (2025)	Real-time ML anomaly detection with dynamic user risk profiling; XGBoost, Random Forest and other classifiers.	Synthetic organizational behavior dataset.	Supports real-time classification into low, medium and high-risk users; XGBoost reports strong performance.	Synthetic data may not fully represent real enterprise telemetry or insider evasion patterns.	Needs validation on benchmark multi-stream insider datasets and integration with cloud-native alerting.
A Comprehensive Survey of XAI Techniques for Malicious Insider Threat Detection (2025)	Survey of ML/DL-based insider threat detection with SHAP, LIME and other explainability techniques.	Multiple ITD datasets and telemetry sources including behavioral, network and psychometric data.	Clarifies how XAI improves analyst trust and feature-level interpretation.	Survey paper; does not implement a complete deployable detection pipeline.	Highlights absence of standard explainability metrics and difficulty combining heterogeneous data streams.
Artificial Intelligence-Based Insider-Threat Detection: A Hybrid Explainable Framework with Automated Response and Privilege Containment (2026)	Hybrid anomaly ensemble using Isolation Forest, PCA reconstruction, Autoencoder and LSTM Autoencoder with SHAP and ARPC.	SEI/CMU CERT Insider Threat Dataset.	Combines static and temporal anomaly signals; adds explainable risk scores and tiered response actions.	Evaluation prioritizes risk ranking and pseudo-label metrics rather than full production validation.	Needs stronger live-cloud implementation and tested policy integration for automated containment.
Insider Threat Detection Model Enhancement Using Hybrid Algorithms between Unsupervised and Supervised Learning (2024)	Unsupervised outlier-score feature augmentation followed by XGBoost supervised classification.	CERT r4.2 dataset.	Improves accuracy to 86.12% with only 20% computing budget; up to 12.5% higher accuracy than comparable methods.	Relies on engineered outlier score matrices and does not emphasize explainable outputs for SOC analysts.	Can be extended with SHAP/LIME explanations and automated Azure ML deployment.
Illuminating the Shadows: An Explainable AI-Driven Approach with Ensemble Learning for Insider Threat Detection (2026)	Random Forest, XGBoost and stacking with temporal feature engineering, SHAP, LIME and LLM-based analyst interpretation.	CERT-based behavioral data with MITRE ATT&CK-aligned filtering.	Dual-layer explainability and SOC-style interpretation improve transparency and alert prioritization.	Full modeling and explainability are applied mainly at weekly granularity due to computational limits.	Needs daily/session-level processing and real-time cloud automation for operational SOC use.

Optimising Insider Threat Prediction: Exploring BiLSTM Networks and Sequential Features (2024)	BiLSTM prediction using manual features, sequential features and previous-day ground truth combinations.	CERT r4.2 dataset.	BiLSTM captures sequential dependencies and improves future-day insider prediction.	Predictive design depends on historical sequences and may need careful tuning for new organizations.	Needs explainability and deployment as a scalable cloud inference pipeline.
BRITD: Behavior Rhythm Insider Threat Detection with Time Awareness and User Adaptation (2024)	Stacked BiLSTM and feedforward neural networks with absolute time encoding and user-adaptive feature sequences.	CMU CERT dataset.	Achieves AUC 0.9730, precision 0.8072 and F1 0.8540, outperforming baselines.	Deep temporal representation may be difficult to interpret without XAI support.	Needs feature-level explanations and deployment-ready monitoring/dashboard components.
BTITD: An Insider Threat Detection Method Based on Behavior-Timestamp Dual-Stream Network (2026)	Dual-stream network: Fourier/CNN behavior stream plus BiLSTM timestamp stream fused with attention.	CERT 4.2 dataset.	Captures multi-scale periodic behavior and long-range temporal dependencies; F1 97.56% binary and macro-F1 94.66% multi-class.	Complex architecture increases implementation cost and does not focus on cloud SOC workflow.	Can be combined with Azure ML, explainability and alert orchestration for practical deployment.
Cloud Intrusion Detection System (2024)	The approach combines Information Gain, Chi-Square, and Particle Swarm Optimization for feature selection, applies SMOTE to fix class imbalance, and uses Random Forest for attack classification.	UNSW-NB15 dataset (trimmed from 49 to 21 features) and the Kyoto dataset (trimmed from 24 to 10 features)	High feature reduction (fewer features needed) leads to faster training times and lower computational overhead compared to deep learning methods, making it ideal for resource-constrained environments.	The performance of the Random Forest classifier can be sensitive to extreme noise or outliers in network traffic.	The study primarily evaluates performance on static, older benchmark datasets (UNSW-NB15, Kyoto). There is a gap in evaluating the system against dynamic, zero-day, or modern contemporary network threats in live cloud environments.
Artificial Intelligence-Based Intrusion Detection System for Cloud Computing (2024)	The approach evaluates machine learning models—including Random Forest, Bayes Network, Random Tree, Naive Bayes, and J48—paired with an Information Gain feature selection strategy using a 10-fold cross-validation scheme.	The research utilized a cleaned and preprocessed subset of the CICIDS2017 dataset, consolidating eight daily traffic files into 2.83 million unique instances to address missing data and class imbalances.	The study demonstrates that feature selection with algorithms like Random Forest and Random Tree achieves high detection accuracy (96.48%) while significantly reducing the execution time and false positive rates.	The paper highlights that minimal weight thresholds for feature selection require expert manual intervention, while certain classifiers like Naive Bayes perform poorly on infiltration traffic and produce undefined precision metrics.	The work leaves open the challenge of automatically identifying optimal feature subsets for specific attack types without reliance on human domain expertise.

Cloud Computing Security Using Intelligent Intrusion Detection Mechanisms (2024)	Implemented a pipeline combining data cleaning, categorical feature engineering, Principal Component Analysis (PCA) for dimensionality reduction, and comparative model evaluation across algorithms like Logistic Regression, KNN, Gaussian Naive Bayes, Linear SVC, Decision Tree, and Random Forest.	The framework collects and preprocesses heterogeneous cloud security inputs, including system logs, network traffic, and SIEM data, using PCA to compress high-dimensional features into a manageable space without losing critical discriminatory information.	The Random Forest Classifier delivers the best overall performance, achieving a test accuracy of 99.88% alongside superior precision and recall compared to traditional rule-based or signature-based detection systems.	The models show minor signs of overfitting during training, while the deployment process carries high computational complexity when scaled up across large cloud infrastructure.	The study relies on a constrained experimental scope that omits testing against zero-day attacks and fails to account for performance variations across different network sizes and complex real-world deployment topologies.
Network Intrusion Detection in Cloud Environments: A Comparative Analysis of Approaches (2024)	he study analyzes signature-based, anomaly-based, behavior-based, and machine learning approaches (e.g., LSTM networks) across public, private, and hybrid cloud deployment models.	As a qualitative review based exclusively on literature sourced via Google Scholar, it synthesizes secondary security data, performance metrics (e.g., detection rates), and compliance costs rather than testing a primary empirical dataset.	The paper highlights that hybrid detection approaches yield the best results—achieving a 95% true positive rate and a low 3% false positive rate—by balancing the known-threat accuracy of signature detection with the novel-threat capabilities of anomaly detection.	Advanced ML and behavior-monitoring solutions present high setup complexity, require extensive datasets, introduce "black-box" interpretability issues, and demand significant computational overhead to handle heavy cloud traffic.	Existing literature lacks sufficient exploration into how quantum computing impacts NIDS efficacy, neglects the socio-technical/human factors in cybersecurity deployment, and provides limited insight into NIDS adaptation for edge/fog computing architectures.
Efficacious Novel Intrusion Detection System for Cloud Computing Environment (2024)	The study proposes a novel Feature Selection (FS) algorithm that modifies the Firefly Algorithm (FFA) using a Decision Tree (DT) classifier as its fitness function, combined with a hybrid classification module featuring Neural Networks (NN) and DTs.	Empirical evaluations were performed using both the CSE-CIC-IDS 2018 benchmark dataset and a custom cloud environment dataset simulated via CloudSim 3.0.3.	The proposed Firefly-Hybrid classifier consistently achieved high-tier performance—reaching precision, accuracy, recall, and F-Measure scores exceeding 0.95 across sample sizes up to 150,000—outperforming standard algorithms like PSO, GA, Random Forest, KNN, and Multi-SVM.	The paper notes potential complexity when scaling to massive environments, as real-time cloud infrastructures are expensive and large-scale parameter reconfiguration remains difficult.	The authors identify a lack of dynamic adaptive systems capable of adjusting to real-time changing cloud locations or resource environments, alongside a need for IDSs that can dynamically scale with cloud Virtual Machines (VMs) and detect novel vulnerabilities.

Adaptive Cloud Intrusion Detection System Based on Pruned Exact Linear Time Technique (2024)	The study proposes the ACIDS-PELT framework, which integrates a Hybrid Harmony Search Optimization and Symmetrical Uncertainty (HSO-SU) feature selection algorithm with the Pruned Exact Linear Time (PELT) change-point detection algorithm and an SVM classifier.	Empirical evaluations were conducted using the standard NSL-KDD benchmark dataset and the real-world ISOT Cloud Intrusion Detection (ISOT-CID) dataset simulated across OpenStack environments.	The proposed framework achieved near-perfect accuracy (99.99%) on NSL-KDD and maintained high accuracy (98.8%) alongside low false positive rates (1.3%–2.9%) after VM scaling on ISOT-CID, outperforming baselines like Random Forest, K-Means, and ACA-TS-IaaS-CD.	The paper notes high computational overhead during continuous stream change-point calculations, potential false alarm fatigue if parameters are uncalibrated, and performance risks against novel, un-captured attack vectors.	The authors address the need for intrusion detection systems that can dynamically adapt to statistical distribution shifts caused by auto-scaling and live VM migrations without relying on change-point approximations.
Intrusion Detection System Using Machine Learning (2024)	The study proposes a hybrid framework that extracts unsupervised features using Non-symmetric Deep Autoencoders (NDAE) and classifies network traffic by embedding Twin Support Vector Machines (TSVM) into a bottom-up merged Decision Tree (DT).	Empirical evaluations were performed using standard benchmark network security datasets, specifically KDD-CUP99 and NSL-KDD.	The proposed NDAE-TSVM-DT framework significantly decreases training time while achieving high levels of accuracy, recall, and precision in classifying attack types (DoS, U2R, R2L, Probe, and Normal).	The paper notes performance dependencies on extensive dataset preprocessing and acknowledges the need for further validation when deploying the deep learning architecture in live, real-time network environments.	The authors address the inability of traditional signature-based and shallow machine learning NIDS to process high volumes of raw traffic efficiently, generalize across varying formats, and adapt to novel attacks without suffering from long training times.

2.1 Individual Research Gap Analysis

Paper 1: Beyond Firewall: Leveraging Machine Learning for Real-Time Insider Threats Identification and User Profiling (2025)

- Existing method: Real-time ML anomaly detection with dynamic user risk profiling; XGBoost, Random Forest and other classifiers.
- Advantages: Supports real-time classification into low, medium and high-risk users; XGBoost reports strong performance.
- Limitations: Synthetic data may not fully represent real enterprise telemetry or insider evasion patterns.
- Research gap: Needs validation on benchmark multi-stream insider datasets and integration with cloud-native alerting.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 2: A Comprehensive Survey of XAI Techniques for Malicious Insider Threat Detection (2025)

- Existing method: Survey of ML/DL-based insider threat detection with SHAP, LIME and other explainability techniques.
- Advantages: Clarifies how XAI improves analyst trust and feature-level interpretation.
- Limitations: Survey paper; does not implement a complete deployable detection pipeline.
- Research gap: Highlights absence of standard explainability metrics and difficulty combining heterogeneous data streams.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 3: Artificial Intelligence-Based Insider-Threat Detection: A Hybrid Explainable Framework with Automated Response and Privilege Containment (2026)

- Existing method: Hybrid anomaly ensemble using Isolation Forest, PCA reconstruction, Autoencoder and LSTM Autoencoder with SHAP and ARPC.
- Advantages: Combines static and temporal anomaly signals; adds explainable risk scores and tiered response actions.
- Limitations: Evaluation prioritizes risk ranking and pseudo-label metrics rather than full production validation.
- Research gap: Needs stronger live-cloud implementation and tested policy integration for automated containment.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 4: Insider Threat Detection Model Enhancement Using Hybrid Algorithms between Unsupervised and Supervised Learning (2024)

- Existing method: Unsupervised outlier-score feature augmentation followed by XGBoost supervised classification.
- Advantages: Improves accuracy to 86.12% with only 20% computing budget; up to 12.5% higher accuracy than comparable methods.
- Limitations: Relies on engineered outlier score matrices and does not emphasize explainable outputs for SOC analysts.
- Research gap: Can be extended with SHAP/LIME explanations and automated Azure ML deployment.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 5: Illuminating the Shadows: An Explainable AI-Driven Approach with Ensemble Learning for Insider Threat Detection (2026)

- Existing method: Random Forest, XGBoost and stacking with temporal feature engineering, SHAP, LIME and LLM-based analyst interpretation.

- Advantages: Dual-layer explainability and SOC-style interpretation improve transparency and alert prioritization.
- Limitations: Full modeling and explainability are applied mainly at weekly granularity due to computational limits.
- Research gap: Needs daily/session-level processing and real-time cloud automation for operational SOC use.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 6: Optimising Insider Threat Prediction: Exploring BiLSTM Networks and Sequential Features (2024)

- Existing method: BiLSTM prediction using manual features, sequential features and previous-day ground truth combinations.
- Advantages: BiLSTM captures sequential dependencies and improves future-day insider prediction.
- Limitations: Predictive design depends on historical sequences and may need careful tuning for new organizations.
- Research gap: Needs explainability and deployment as a scalable cloud inference pipeline.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 7: BRITD: Behavior Rhythm Insider Threat Detection with Time Awareness and User Adaptation (2024)

- Existing method: Stacked BiLSTM and feedforward neural networks with absolute time encoding and user-adaptive feature sequences.
- Advantages: Achieves AUC 0.9730, precision 0.8072 and F1 0.8540, outperforming baselines.
- Limitations: Deep temporal representation may be difficult to interpret without XAI support.
- Research gap: Needs feature-level explanations and deployment-ready monitoring/dashboard components.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 8: BTITD: An Insider Threat Detection Method Based on Behavior-Timestamp Dual-Stream Network (2026)

- Existing method: Dual-stream network: Fourier/CNN behavior stream plus BiLSTM timestamp stream fused with attention.
- Advantages: Captures multi-scale periodic behavior and long-range temporal dependencies; F1 97.56% binary and macro-F1 94.66% multi-class.
- Limitations: Complex architecture increases implementation cost and does not focus on cloud SOC workflow.
- Research gap: Can be combined with Azure ML, explainability and alert orchestration for practical deployment.
- Possible improvement: Use Azure-based ingestion, model hosting, SHAP explanations, alerting and dashboard visualization to make the detection workflow more operational.

Paper 9: Cloud Intrusion Detection System (2024)

- Existing method: Hybrid cloud IDS using Synthetic Minority Over-sampling Technique (SMOTE) for class balance, a hybrid feature selection pipeline (Information Gain, Chi-Square, and Particle Swarm Optimization), and a Random Forest classifier tested on UNSW-NB15 and Kyoto datasets.
- Advantages: Effectively mitigates target class imbalance; drastically reduces feature counts (from 49 to 21 on UNSW-NB15, and 24 to 10 on Kyoto) while yielding accuracy above 98% and 99%, respectively.
- Limitations: High computational overhead during the offline training phase due to combined bio-inspired population search (PSO) and iterative SMOTE sampling.
- Research gap: Lacks real-time model updating and edge-level optimization mechanisms to sustain high accuracy under shifting network traffic conditions.
- Possible improvement: Introduce distributed multi-objective metaheuristics with adaptive parameters and apply model compression techniques (such as model pruning or quantization) for efficient deployment on resource-constrained cloud/edge nodes.

Paper 10: Artificial Intelligence-Based Intrusion Detection System for Cloud Computing (2024)

- Existing method: Data pre-processing and Information Gain (IG) feature selection evaluated across multiple classifiers (Random Forest, Bayes Network, Random Tree, Naive Bayes, and J48) using the benchmark CICIDS2017 dataset.
- Advantages: Drastically reduces dataset complexity by pruning features down to just 4 critical attributes while maintaining high detection accuracy (96.48% with RF and RT) and identifying Naive Bayes as highly effective for DoS/DDoS detection.
- Limitations: Severe data loss during cleaning (discarding missing/unlabeled data instances) and vulnerability to performance degradation on unbalanced non-DoS attack classes when using hyper-reduced feature subsets.
- Research gap: Lacks robust data imputation and class-rebalancing techniques to preserve subtle signature patterns for rare attack types in heavily pruned datasets.
- Possible improvement: Combine Information Gain with synthetic oversampling techniques (e.g., SMOTE) or deep autoencoders to prevent information loss and improve minority class detection rates.

Paper 11: Cloud Computing Security Using Intelligent Intrusion Detection Mechanisms (Vol. 3, No. 1)

- Existing method: Principal Component Analysis (PCA) for feature dimensionality reduction combined with multiple machine learning classifiers (Logistic Regression, K-Neighbors, Gaussian Naive Bayes, Linear SVC, Decision Tree, and Random Forest).

- Advantages: Achieves high performance using Random Forest (99.88% test accuracy), while PCA integration significantly reduces computational overhead and training time while maintaining high accuracy (99.82%).
- Limitations: Focuses on standard static machine learning evaluation without testing resilience against adversarial evasion or zero-day attack variants.
- Research gap: Fails to address continuous stream learning or dynamic model retraining in response to evolving zero-day threat patterns.
- Possible improvement: Integrate dynamic online learning models paired with autoencoders for non-linear dimensionality reduction to capture complex, non-linear cloud traffic patterns better than standard linear PCA.

Paper 12: Network Intrusion Detection in Cloud Environments: A Comparative Analysis of Approaches (2024)

- Existing method: Qualitative research design using thematic literature analysis to compare traditional signature/anomaly-based mechanisms with modern machine learning, user behavior monitoring, and cloud-native containerized detection approaches.
- Advantages: Demonstrates that hybrid detection methods achieve superior performance compared to single-approach systems, reaching a 95% True Positive Rate (TPR) and a 3% False Positive Rate (FPR).
- Limitations: Heavy reliance on qualitative literature review methodologies rather than original empirical benchmarking on unified datasets; anomaly-based components still exhibit high false positive rates (8%).
- Research gap: Lacks a single unified quantitative evaluation framework to benchmark privacy scores alongside network latency across dynamic multi-cloud environments.
- Possible improvement: Implement automated real-time feature selection frameworks to dynamic multi-cloud deployments, balancing privacy score requirements with lower detection latency.

Paper 13: Efficacious Novel Intrusion Detection System for Cloud Computing Environment (2024)

- Existing method: Hybrid cloud IDS combining a Modified Firefly Algorithm (FA) with a Decision Tree wrapper for feature selection, followed by a coupled Neural Network and Decision Tree (NN-DT) ensemble.
- Advantages: Demonstrates strong classification performance on modern datasets (CSE-CIC-IDS2018) by combining rule-based decision logic with neural pattern recognition.
- Limitations: Firefly Algorithm risks premature convergence in high-dimensional feature spaces; wrapper-based feature selection incurs high offline training latency; static model with no online drift adaptation.
- Research gap: Lacks automated mechanisms to adapt to concept drift over time, and suffers from computational complexity that hinders deployment in resource-constrained edge/cloud nodes.

- Possible improvement: Introduce multi-objective metaheuristics with adaptive parameters to avoid premature local optima, and integrate lightweight model compression (e.g., pruning/quantization) for real-time edge-cloud deployment.

Paper 14: Adaptive Cloud Intrusion Detection System Based on Pruned Exact Linear Time Technique (2024)

- Existing method: Adaptive cloud IDS (ACIDS-PELT) using Harmony Search Optimization with Symmetrical Uncertainty (HSO-SU) for feature selection, Pruned Exact Linear Time (PELT) for change-point detection, and SVM for classification.
- Advantages: Dynamically detects statistical change-points in cloud traffic profiles to trigger automated model retraining during network shifts or auto-scaling.
- Limitations: High SVM retraining overhead $\mathcal{O}(N^2)$ during frequent change-points; static initial feature subset fails to adapt when new attack types emerge; potential false-positive change triggers from non-malicious traffic spikes.
- Research gap: Lacks continuous dynamic feature space re-selection during online drift and requires real-time integration into live cloud orchestrators (e.g., Kubernetes/OpenStack telemetry).
- Possible improvement: Replace standard SVM with incremental/online learning classifiers (e.g., Online Random Forests or SGD) and implement dynamic feature importance updates during change-point events.

Paper 15: Intrusion Detection System Using Machine Learning (2024)

- Existing method: Hybrid classification framework using Decision Trees (DT) and Twin Support Vector Machines (TSVM) with a bottom-up merging structure, alongside protocol-based filtering.
- Advantages: TSVM offers faster computational speed compared to standard SVMs; the bottom-up tree structure helps mitigate error accumulation across decision nodes.
- Limitations: Conceptual mismatch in methodology (mentions deep autoencoders in objectives but uses classical ML in architecture); relies on obsolete datasets (KDD-Cup 99, NSL-KDD); lacks concrete performance baseline metrics.
- Research gap: Needs evaluation against modern cloud network traffic datasets and rigorous empirical validation comparing inference latency and false-alarm rates.
- Possible improvement: Standardize the architecture by integrating actual deep autoencoder feature extractors with light gradient boosting models, and evaluate on modern multi-class telemetry.

3 Project Objectives

- Design a cloud-based insider threat prediction framework using multi-stream user activity logs.
- Preprocess CERT r4.2 logs into session-level behavioral features for machine learning.

- Train and compare ensemble and sequential models for accurate insider threat prediction.
- Integrate SHAP-based Explainable AI to provide feature-level justification for each risk score.
- Plan a secure Azure architecture for storage, processing, authentication, monitoring and visualization.
- Provide real-time alerting and dashboard support for SOC-style investigation.

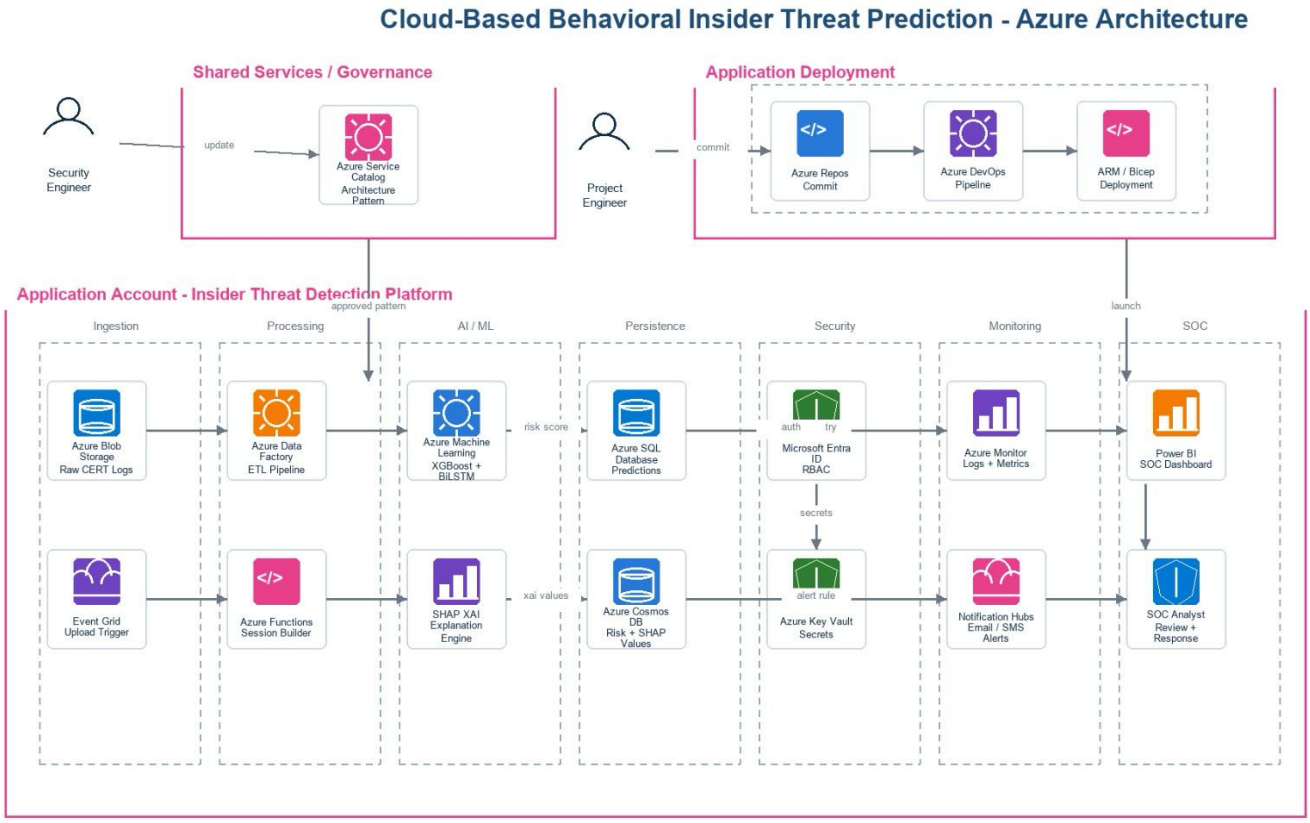
4 Novelty Summary

- Combines behavioral profiling with temporal prediction so that both sudden anomalies and gradual insider behavior changes can be detected.
- Adds explainable risk scoring using SHAP, allowing analysts to see which actions influenced each alert.
- Uses Azure services to convert offline research ideas into a scalable cloud workflow with automated ingestion, monitoring and visualization.
- Links prediction results to a SOC-friendly workflow through Power BI dashboards and alert notifications.
- Focuses on secure role-based access using Microsoft Entra ID and monitored execution through Azure Monitor.

5 Proposed Architecture

5.1 Azure Cloud Architecture Diagram

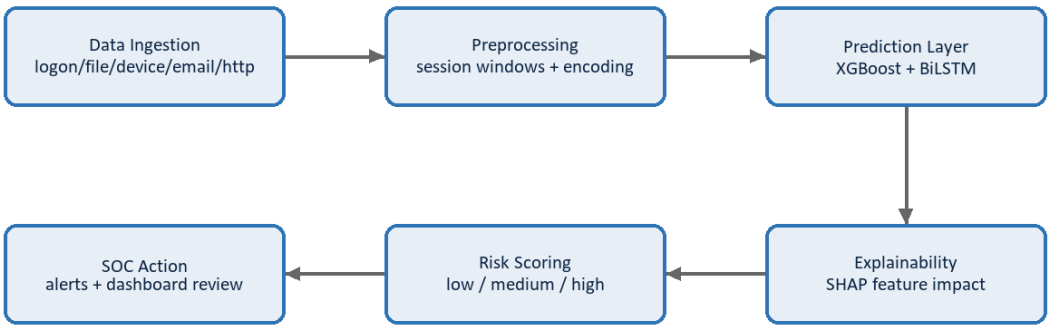
The Azure architecture shows how raw user activity logs move through storage, processing, model inference, explainability, database storage, monitoring and dashboard layers.



5.2 Complete System Architecture Diagram

The complete system workflow starts from multi-source log ingestion and ends with risk-based SOC action.

Diagram 2: Complete System Workflow



6 Dataset Details

Dataset Name	CMU CERT Insider Threat Dataset r4.2
Source	Carnegie Mellon University Software Engineering Institute / CERT Insider Threat Center
URL	https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=508099 and Kaggle mirrors for r4.2 access
Approximate Size	About 1.5 GB compressed and about 13 GB after extraction, depending on distribution source
Records	Millions of timestamped user activity events across employees and scenarios
Raw Streams	logon.csv, file.csv, device.csv, email.csv, http.csv and related user metadata
Data Type	Temporal structured CSV event logs
License	Research/academic benchmark dataset; usage must follow the source distribution terms
Purpose	Train and evaluate behavioral anomaly detection and insider threat prediction models
Preprocessing Required	Timestamp parsing, user-session aggregation, categorical encoding, class balancing, feature scaling and train-test splitting

7 Azure Services Planning

Azure Service	Purpose in the Project
Azure Blob Storage	Stores raw CERT logs, processed feature files and exported model artifacts.
Azure Data Factory	Coordinates ETL pipelines for cleaning, transforming and aggregating activity logs.

Azure Functions	Runs event-driven preprocessing or alert logic when new files or predictions arrive.
Azure Machine Learning	Trains, evaluates and deploys XGBoost/BiLSTM models and SHAP explainability jobs.
Azure SQL Database / Cosmos DB	Stores prediction records, user risk scores, alert status and explanation values.
Microsoft Entra ID	Provides authentication and role-based access control for users and services.
Azure Monitor	Tracks service health, model endpoint logs, function execution and alert metrics.
Azure Notification Hubs	Sends notification alerts for high-risk insider sessions.
Power BI	Displays SOC dashboards with risk trends, user profiles and SHAP feature contribution charts.
Azure Key Vault	Protects secrets, connection strings and API keys used by the pipeline.

8 Implementation Status and Contribution

The current contribution is focused on individual literature survey and project planning.

Activity	Contributor	Status
Literature survey of assigned 8 papers	Rahul R (23BIT0096) Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1
Individual research gap analysis	Rahul R (23BIT0096) Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1
Dataset selection and preprocessing plan	Rahul R (23BIT0096) Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1
Azure cloud architecture planning	Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1

XAI-based model workflow planning	Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1
Review-1 report documentation	Rahul R (23BIT0096) Nitheeshwar S (23BIT0302)	Completed / Planned for Review-1

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